

Heartland: A Kidney, Liver, Heart Disease Patient Assistance Smartphone Application With Disease Detection, Treatment, Physician Suggestion Features

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Abstract - Any problems with the liver, kidney, and heart Can have serious repercussions and increase mortality if neglected. An online healthcare assistance system for the detection of liver, kidney, and heart disease was not presented in the previous works. Features like checking medication information, scheduling physician appointments and online tests, checking vital signs, searching for kidney donors, sharing daily information with medical professionals, treatment, and searching for disease diagnosis centers were missing in previous works. To resolve existing constraints, this paper presents a healthcare assistance mobile application that includes features for detecting liver, kidney, and stroke disorders, suggesting treatments, scheduling labs and physician tests, searching for diagnosis centers, sharing daily information with doctors, verifying the authenticity of medication information, checking vital signs-based health status, and online-based healthcare product purchases. According to the app's competency and efficiency results, over 59% of reviewers left positive comments on the application.

Keywords— *Diagnosis Center Search, Heart Disease, Kidney Disease, Liver Disease, Physician Search, Smartphone App.*

I. INTRODUCTION

Currently, the number of people who suffer from and die from stroke, kidney, liver, and heart disease is increasing due to changes in lifestyle, junk food consumption, and physical inactivity. Approximately 9 million individuals died from heart disease in 2019, compared to 2 million in 2000. According to WHO, stroke, a condition in which the brain's blood flow is insufficient and causes cell death, killed 1.34 lakh individuals in Bangladesh last year, or 82.3 deaths per 1 lakh. A stroke is a medical condition that causes a partial loss of brain cells. Neurologists claim that if a patient is admitted to the hospital during the first few hours after a stroke, they can recover completely on medicine alone [1]. Strokes are more common in men and the elderly, and their risk increases with age. In Bangladesh, lack of physical activity, high weight, diabetes disorder, alcoholism, smoking, hypertension, and dyslipidemia are the most prevalent risk factors for stroke. Maintaining a healthy lifestyle, lowering blood pressure, stopping smoking,

exercising, being checked for diabetes, eating healthily, and getting medical help can all help people avoid stroke [2]. A stroke patient may have headache, vomiting, sudden fainting, difficulty or inability to speak, unresponsiveness of specific body part, and light-headedness in addition to severe pain in the head, neck, face, and occasionally between the eyes [3]. In such cases, the doctor must do a comprehensive assessment and offer suitable medicine and rehabilitation guidance. In Bangladesh, one in three persons suffers from liver disease, yet there is only one hepatologist for every two million people. Fatty liver disease affects more than 40 million people, according to [4]. Hepatitis B affects about 10% of the country's population. Among Bangladeshis, hepatitis is common along with fatty liver disease. Vaccination is a very effective method of preventing liver diseases and hepatitis. For patients with liver problems, liver transplants are an additional alternative.

According to [4], many liver disease sufferers do not get online healthcare platforms that offer features including disease detection, lab test application, online meeting scheduling, and consultant search, and treatments suggestion. An estimated 38 million people in Bangladesh suffer from kidney disease, which is believed to be the cause of 40 thousand fatalities annually. The number of kidney disease patients in the country has skyrocketed in recent years. Diabetes and hypertension are the two main causes of kidney damage [5]. Dialysis, medication, and kidney transplantation are some treatments for severe kidney failure. However, kidney patients currently suffer greatly due to the lack of online ICT-based facilities for kidney donor and consultant searches, and treatment help systems. There are now a number of studies on the support of kidney, liver, and stroke diseases. The study in [6] created a smart phone application for people with chronic illnesses that predict Parkinson's, skin, and cardiovascular diseases using an RNN-based DL model. An Arduino Uno, pulse, glucose, ECG, and pressure sensors were used in [7] to create a real-time chronic renal patient health data collection system. Both gradient boosting and LSTM algorithms were used by the authors in [8] to predict the renal function status of

kidney patients. An Android mobile application with search and donation capabilities for blood donors and platelets was created by the article in [9]. A smart phone application that provides user involvement, instant messaging, and awareness program features for diabetes preventive issues was produced by the article in [10]. The authors in [11] use a feed forward neural network model to forecast the mortality risk of liver transplant recipients. Using DL and NLP technology, the study in [12] created a system for recommending meals, physical activity, and cholesterol risk detection. The effectiveness of several AR and VR games for the post-stroke healing process was examined in the work in [13]. Existing literature did not assess users' responses to queries about skin, urine, chest, blood, injury, body, various test results, and disease symptoms in order to identify the risk of stroke disorder, kidney disease, and liver disease. There isn't any online support system in the literature that verifies the legitimacy of medications, gives information about diagnosis centers, or allows doctors to plan appointments.

The existing works lack kidney donor search capabilities and kidney donation features. There were no online lab-based health test application features, online health care-related e-marketplace features, or features for assessing the status of vital signs in the earlier works. No support system that offered daily patient information and activity sharing capabilities with the doctor was available in the existing system. Existing works that deal with diagnosis and treatment information access features for liver disease, renal disease, diabetes, and stroke disorders need patient aid systems. To outperform the existing issues, this paper presents a smart phone application for healthcare that uses React Native and Firebase technologies to help identify and diagnose kidney, stroke, and liver diseases. The foremost and useful contributions of this article are proffered below:

- (i) By examining patients' responses on their symptoms, health issues, and test results, the proposed "Heart Land" mobile application determines their risk of developing heart disease, kidney disease, diabetes, and liver disease. By tracking user data and preferences, this smart phone application also provides online meeting scheduling tools and a physician search features for individuals with kidney, liver, and stroke conditions.
- (ii) By determining the patient's disease name and needs, the proposed patient assistance mobile application also offers a feature that verifies the validity of medication information and a search function for the patient's health diagnosis centre. This mobile application offers kidney donor registration, online healthcare test application filing, patient status checking based on vital sign results, and daily information exchange between patients and doctors.
- (iii) Based on the patient's symptoms, the healthcare assistance smart phone application suggests treatment and diagnosis options for stroke, liver, and kidney patients. Additionally, this mobile application offers options for purchasing healthcare products based on user preferences and payment.

Based on patient evaluation comments, this article also presents app features competency and helpful impact analysis results.

In section two, the literature review is presented. Section 3 elaborates on the design and descriptions of the suggested mobile application. Section four shows the results of the app competency evaluation. Section 5 discusses the last remarks for proposed work as well as a few research challenges.

II. LITERATURE REVIEW

This section provides a summary of the literature on the kidney, liver, and stroke disorders assistance. A mobile application for emergency medical care was created by the article in [14] and offers features including ambulance calling capabilities and patient data access via QR codes. Using OCR and NLP technology, the work in [15] provided the renal patient with a personalized chatbot service, patient classification, and medical report scanning capabilities. For patients with nutritional disorder, the paper in [16] created an Android application that provides DNN-based meal identification features. Using the ML and Sarima model, the study in [17] examined the prevalence of kidney illness in Sri Lanka. The pH color of kidney patients was identified by the authors in [18] using a different supervised learning technique.

The study in [19] used a 3D ultrasound Doppler imaging equipment to track vascular changes in individuals with kidney disease. The random forest technique was employed by the authors in [20] to forecast medicine dosages for patients undergoing kidney transplantation. Using cloud computing technology, the work in [21] created a web application for TB patients that offer features including TB treatment, doctor time optimization, and healthcare provider support services. In [22], the authors introduced a smart phone app for patients with renal disease that offers capabilities for recording medical test results, learning about kidney disease, and collecting information on past patient histories. The study in [23] employed SVM classifiers and ultrasound pictures to identify kidney abnormalities. Using CT scans, CNN algorithms, and Internet of Things sensors, the authors in [24] forecast the state of a healthy liver.

The work in [25] uses PSP Net techniques and encoder-decoder networks to segregate liver tumors. The random forest model was used in the article in [26] to predict the patients' hepatitis disorder status. The study in [27] predicted the patient's survival status following liver transplant surgery using both the BiLSTM and RNN models. Using evolutionary algorithms, the authors in [28] created a customized meal advice system for kidney failure patients. A mobile application for diabetes and cardiovascular disease prevention and awareness was shown in [29].

Unlike other works, this article provides a smart phone application for kidney, liver, and stroke patients that include features like disease detection, doctor search, online meeting schedule, lab test application, kidney donor search, medication verification, and treatment suggestion.

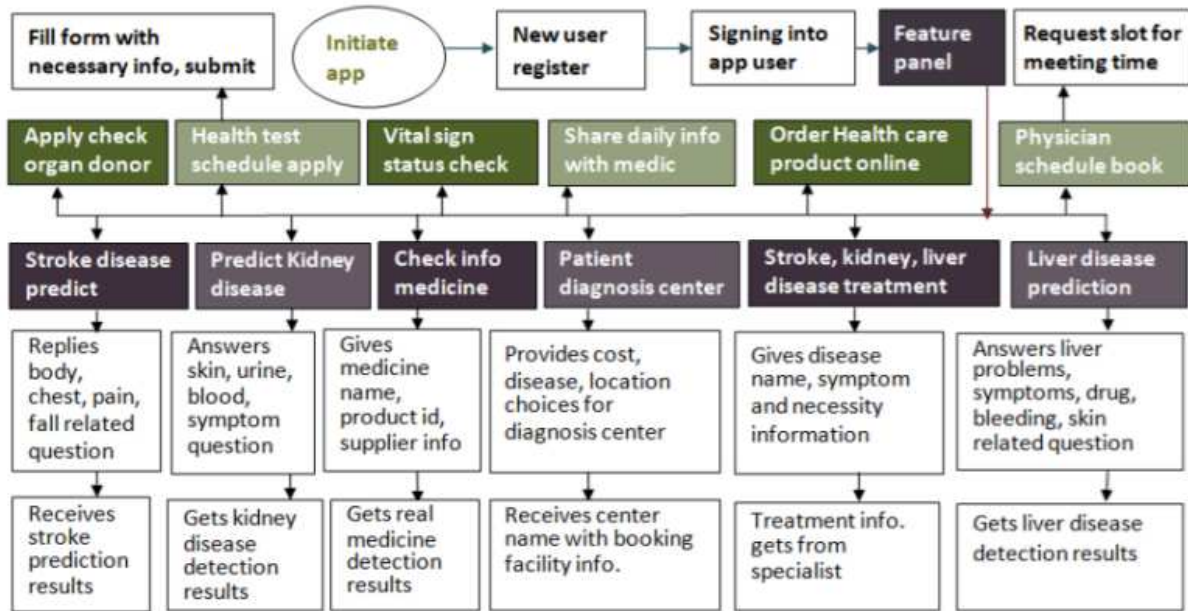


Fig. 1. Proposed kidney and heart disease assist app flowchart

III. PROPOSED SCHEME

Figure 1 dispatches the flowchart model for the smart phone application (“Heart Land”) that helps people with liver illness, stroke, and Kidney’s disease. This app only screens disease and not used for real time prediction. The app name heartland is used once only in title and no validity are given. Reactive Native and Firebase are used for front-end and back-end development, respectively.

A. Kidney, stroke, liver disease detection, medical search

Using their account name, email address, national ID card number, and user verification code, users can complete the app registration form (Figure 2(a)). Using their name and verification code, the user can complete the app user signing completion form (Figure 2(b)). Following appropriate signing action, the app features panel is accessible (Figure 2(c)). By responding to questions related to diseases and illness, the user can identify their risk of heart disease and stroke (Figure 2(d)). By responding to questions on blood pressure, swelling, nausea, vomiting, urine problems, and skin problems, the patient can determine their risk of kidney disease (Figure 2(e)). Using manufacturing ID, and supplier name, the patient can verify the medication (Figure 2(f)). By entering the cost, category, and location, the patient can get disease diagnosis centre information (Figure 2(g)).

By responding to questions about thirst, hunger, weight, fatigue, urination, vision, and infections, the user can determine their risk of developing diabetes (Figure 2(h)). By responding to questions on jaundice, abdominal pain, swelling, bruises, itchy skin, and various liver-related test results, the patient can determine the status of liver illness (see Figure 2(i)).

B. Physician, health test schedule book, vital sign detect

By providing information about cost, expertise, and location, the patient can look for the appropriate medical term for the treatment of their illness (Figure 2(j)). By providing the meeting date, time, payment, user contact, and chamber name, the patient can plan an online appointment with a chosen physician (Figure 2(k)). By providing the organ name, associated medical conditions, and contact details, the patient can submit an organ donation request (Figure 2(l)). By providing blood type, contact details, and other test results, the patient can look up the name of the kidney donor (Figure 2(m)).

Along with requesting a meeting time, the patient can apply online for any necessary lab tests relevant to their health (Figure 2(n)). Figure 2(o) displays the patient’s health state and allows them to provide various blood pressure, glucose, heart rate, weight, and body temperature data.



Fig. 2. (a) app register steps, (b) signing process, (c) features panel, (d) heart disease risk detection, (e) kidney disease risk detection, (f) medicine information checking, (g) diagnosis centre search, (h) diabetes disorder risk detection, (i) liver disease risk detection, (j) physician search, (k) physician schedule booking, (l) organ donor apply, (m) kidney donor search, (n) health test schedule apply, (o) vital sign

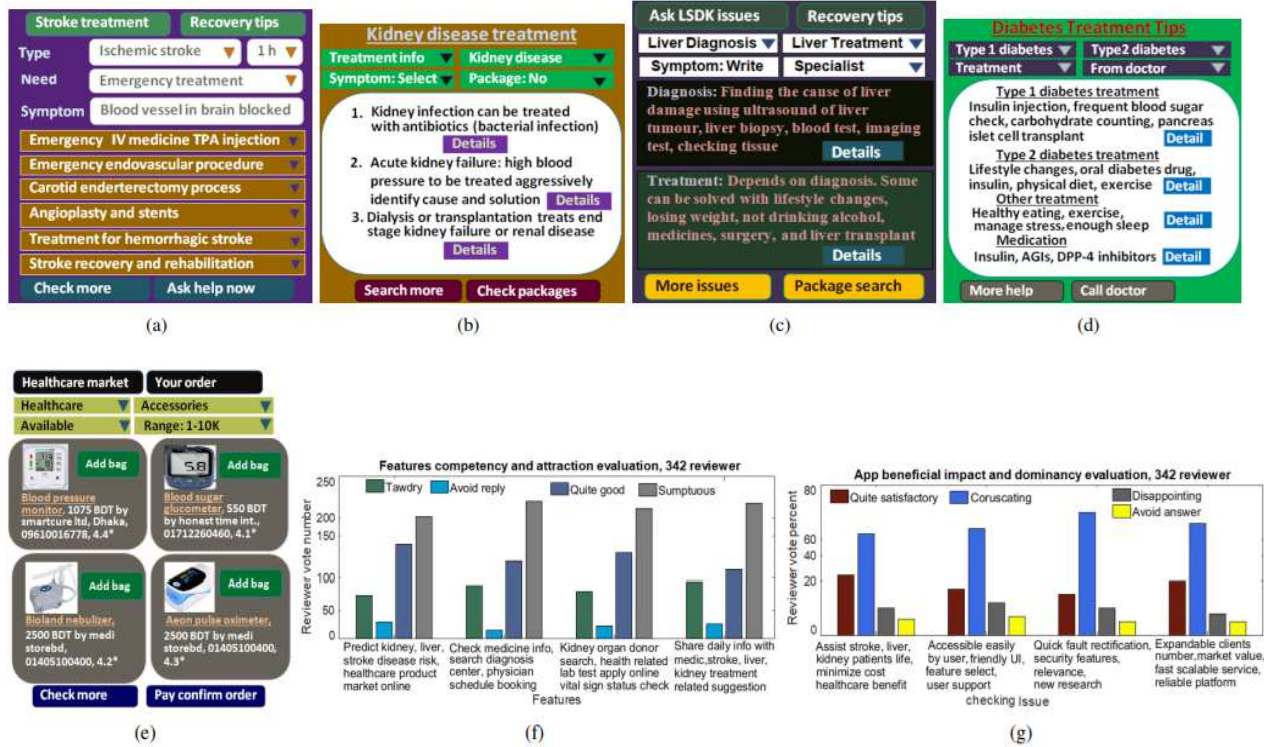


Fig. 3. (a) Stroke treatment suggestion, (b) kidney disorder treatment suggestion, (c) Liver diagnosis and treatment suggestion, (d) diabetes treatment tips, (e) health care product e-market, (f) features competency evaluation results, (g) app beneficial impact evaluation results

TABLE I
TABLE OF PERFORMANCE COMPARISON

Work Name	Prediction of stroke, kidney, liver, diabetes disease	Physician schedule book, medicine, vital sign check	Kidney donor search, health test schedule apply	Share info with medic, disease treatment suggestions
Rodriguez et al., [10]	Not propounded	Not propounded	Not propounded	Not propounded
Lum et al., [18]	Not propounded	Not propounded	Not propounded	Not propounded
Panda et al., [20]	Not propounded	Not propounded	Not propounded	Not propounded
Sihananto et al., [28]	Not propounded	Not propounded	Not propounded	Not propounded
Our App	Propounded	Propounded	Propounded	Propounded

C. Share information with medic, treatment suggestion

By sharing information on daily food consumption, medications, symptoms, and exercise, the patient can obtain the doctor's perspective regarding daily progress (Fig. 2(p)). By providing stroke type, the user can receive therapy suggestions (Fig. 3(a)). By providing the name of the symptom, the patient can obtain information on various renal disease treatment packages (Fig. 3(b)). The user can use the screen in Figure 3(c) to view the information about the liver disease diagnosis and treatment package. The screen in Figure 3(d) allows access to several diabetes kinds along with treatment details. The patient can use the screen shown in Figure 3(e) to choose, order, and buy any healthcare product online.

IV. RESULTS AND ANALYSIS

In this section, 342 reviewers' comments are evaluated in order to examine the app. The results of the features competency and attraction evaluation are shown in Figure 3(f). Reviewers' sumptuous, quite good, tawdry, and avoid responding remarks for the app's mentioned features ranged from 202 to 220, 110 to 140, 80 to 96, and 10 to 20, respectively, according to Figure 3(f).

Figure 3(g) showed that users' coruscating, quite satisfactory, disappointing, and avoid answer reply percentages ranged from 62-70%, 15-22%, 8- 12%, and 5-7%, respectively, for all examined app benefits and dominant issues. In contrast to previous works in Table 1, this app offers new capabilities like the ability to detect stroke, liver, and kidney problems, suggest treatments, organize meetings for lab and physician tests, check vital signs, and communicate with medical professionals.

V. CONCLUSION

This article contributes a smart phone application for chronic illness support that estimates the risk of liver, kidney, and stroke disorders based on the patient's symptoms, medical conditions, and test results. For patients with renal, liver, and heart problems, the suggested smart phone application offers capabilities for doctor search, medication information authenticity verification, online doctor appointment scheduling, and diagnosis centre search. Users of the suggested healthcare mobile application can look for kidney donors, order medical tests online, and check the state of their vital signs. According to the app's competency and helpful impact evaluation results, over 59% and 62% of users provide sumptuous and coruscating comments regarding the proposed healthcare application. IoT and deep learning based vital sign status monitoring, block chain and cloud computing based patient data authentication and access facilities, DL and explainable AI based heart, liver, and kidney disease reason detection, ML and AI based treatment suggestion, are some of the future research challenges.

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